

Luigi Pennisi

Gameplay Engineer | UE5 · C++ · Godot

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Profiles

 **Github**
[Ruy41321](#)

 **LinkedIn**
[luigipennisi](#)

 **itch.io**
[Ruy41321](#)

Summary

Gameplay Engineer with a strong foundation in **C++** and **Unreal Engine 5**. After completing the rigorous **42 Roma Common Core**, I honed advanced problem-solving and communication skills, alongside a specialized skill set in low-level memory management and high-performance system architecture. Experienced in game development through independent projects where I acted as both **Game Programmer** and **Lead Producer**, managing the technical implementation of modular frameworks such as the Gameplay Ability System (GAS) while coordinating team efforts. I apply a deep technical interest in software design patterns to engineer **scalable systems** for immersive, **creative, and engaging games**.

Main Skills

 **C / C++ Programming**
Advanced
Modern C++ (17/20), Multithreading, Memory Profiling

 **Unreal Engine 5**
Advanced
C++, UE Blueprints, GAS, Replication, Perforce

 **3D Math & Geometry**
Advanced
Raycasting, DDA, Trigonometry, Linear Algebra

Secondary Skills

 **Software Engineering**
Advanced
Perforce, CI/CD Pipeline, Git, Design Patterns, Docker, Linux, Python, Java

 **Godot 4**
Intermediate
GDScript, 2D, AI Programming, Replication

 **Full Stack Web Technologies**
Advanced
Vue.js, PHP, Websockets, JSP, Javascript

Unreal Engine Projects

Makhia January 2026 - Present
Online 1v1 Melee RPG | UE 5.7, C++, GAS, EOS

- **GAS Multiplayer Architecture:** Implemented a robust combat framework leveraging GAS to handle **attribute replication, server-side validation, and client-side prediction**, ensuring responsive mechanics.
- **EOS P2P Integration:** Integrated **Epic Online Services** for Peer-to-Peer connectivity and matchmaking, ensuring long-term game availability and zero server overhead costs.
- **MVC Pattern Implementation:** Architected a clear **MVC separation** by implementing the **Model in C++** for core logic and state management, while utilizing **UE Blueprints for the View** and UI feedback.
- **Modular Architecture:** Engineered a decoupled Character System using **C++ base classes** and Data Assets for scalable RPG mechanics, including a **custom C++ locomotion and movement system** developed to replace standard Blueprint templates for enhanced control and performance.

TargetShootingGame July 2025
Simple FPS | UE 5.6, Nanite, Lumen, Chaos Physics

- As my first UE project, it served as an introduction to several key **UE5** concepts.

Other Projects

Cub3D

Developed in C from Scratch | 3D | Raycasting | DDA

November 2024 - March 2025

IceSkating

Godot 4 | Multiplayer | 2D Puzzle Game | Procedural Maps

June 2025 - July 2025

Web Pong

Web-App | Multiplayer Pong Game | Real-time using WebSockets | VueJS | FastAPI

January 2026 - April 2026

Work Experience

KeyBiz

Freelance Game Developer

Remote

11/09/2025 - 10/10/2025

- Developed a web-based mini-game using **Godot**, optimized for high-traffic corporate websites.
- Ensured cross-device compatibility with a focus on **mobile responsiveness** and performance.
- Built a modular system to facilitate **easy level expansion** and content updates.

Magnus Monitor

Intern

Galway, IRL

01/10/2022 - 31/12/2022

Assembly and configuration of **industrial IoT devices** for liquid tank monitoring.

Education

42 Rome

Software Engineering & Advanced Computer Science

Common Core Curriculum • (Level 10 - Completed)

Rome, IT • 08/01/2024 - 14/04/2026

Completed the rigorous, project-based **42 Rome Common Core**, mastering **C/C++**, memory management, and network protocols. Acquired strong **problem-solving** skills by delivering complex software solutions autonomously within a peer-to-peer environment.

Currently pursuing **Advanced Specialization** in [Graphics & Gaming, Algo & AI & Data, Web & Mobile]

ITI B. Lucarelli

Information Technology & Telecommunications

High School Diploma • 100/100

Benevento, IT • 01/10/2019 - 08/07/2022

- Learned to program in C/C++, Java, JavaScript and PHP.
- Developed production-ready Web & Mobile Applications.

Certifications

CCNav7: Introduction to Networks

Cisco Networking Academy

27/05/2022

Introduction to Cybersecurity

Cisco Networking Academy

18/12/2020

IT Essentials

Cisco Networking Academy

29/01/2022

Languages

English

Full Professional Proficiency



Italian

Native



Interests

🎮 Passionate Gamer

MMO, RPG, FPS, Survival